

II-Probabilistic Approaches Inference

Foundations of Neuro-Symbolic AI

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Outline

- ▶ Variable Elimination
- ▶ Approximative inference

Contraction by sub-contractions

Idea: Close one variable after another in a contraction.

Tree width

Size of the largest cluster is the "bottleneck".

Hardness of contractions

Finding the best order of the variables to be contracted is NP-hard, since related to the triangulation problem of a graph.

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